

Jad Akil

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[GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Motivated Computer Science graduate with a strong foundation in Artificial Intelligence and Machine Learning Experienced in building and optimizing ML/DL models using TensorFlow and PyTorch.

Skilled in backend development, distributed systems, and cloud technologies.

Passionate about designing scalable systems, solving complex problems, and applying AI in real-world scenarios.

EDUCATION

Lebanese University (LU) — Bachelor's Degree in Computer Science
Lebanon | **Graduated: 2025**

- Specialized in Python programming, Machine Learning, and Software Development
- Strong foundation in algorithms, data structures, and system design
- Focused on practical AI applications, data analysis, and problem-solving

TECHNICAL SKILLS

Programming Languages:

Python (AI/ML/Deep Learning), Java (OOP, Design Patterns, Android), C/C++ (Data Structures & Algorithms), JavaScript (Basic Mern Stack).

AI & Machine Learning:

scikit-learn, TensorFlow, PyTorch, Neural Networks, Data Preprocessing, NLP Basics

Backend & APIs:

FastAPI, Node.js, Express.js, RESTful APIs, JWT, OAuth Authentication

Databases:

MongoDB, Oracle ,vector database (Indexing, Query Optimization, Performance Tuning)

Distributed Systems & Big Data:

Parallel Programming (OpenMP, MPI), Distributed Systems Fundamentals, Hadoop, Spark system

Cloud & Certifications:

AWS Cloud Practitioner Certified ([CLF-C02](#))

Tools : Git, GitHub, VS Code, Postman .

PROJECT EXPERIENCE

Machine Learning Project – Heart Disease Prediction ([Framingham](#))

- Developed a Decision Tree model to predict 10-year coronary heart disease (CHD) risk
 - Improved model performance from baseline to ~70% accuracy through data preprocessing, balancing, and hyperparameter tuning
 - Achieved **80% accuracy on synthetic patient testing (8/10 correct predictions)**
 - Applied **GridSearchCV (5-fold cross-validation)** and pruning to optimize model performance
 - Engineered key features contributing to ~90% of model decisions (blood pressure, age, smoking)
 - Addressed class imbalance using downsampling after evaluating SMOTE and class weighting techniques
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Chess Hybrid AI Platform ([Full-Stack Project](#))

- Developed a full-stack chess engine combining a scalable Python backend with a structured machine learning training pipeline React frontend with FastAPI-based backend communication
 - Designed and implemented a **shard-based data pipeline**, enabling efficient processing of millions of samples without exceeding memory limits
 - Built a PyTorch-based **policy + value neural network** to predict moves and evaluate board positions
 - Engineered a **streaming data loader**, avoiding full dataset loading and improving training scalability
 - Implemented data preprocessing systems including **deduplication, validation, and filtering** to ensure high-quality training data
 - Designed a modular training architecture with **replay buffer, iterative training, and validation tracking**
 - Added checkpointing system for saving and comparing model performance across training iterations
 - Structured the backend into clear components (data processing, model, training), improving maintainability and extensibility
 - Focused on performance and memory efficiency, simulating real-world ML system constraints
 - Built the system to support future extensions such as **self-play reinforcement learning and MCTS integration**
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My_Restaurant — Android App ([Java](#))

- Developed a native Android restaurant application using MVC architecture
- Managed menus, orders, and table tracking with clear separation of concerns
- Designed for scalability and future features such as payments and reservations

LANGUAGES : Arabic: Native / English: Intermediate